

ZS571KL

Board-Level Trouble Shooting Guide

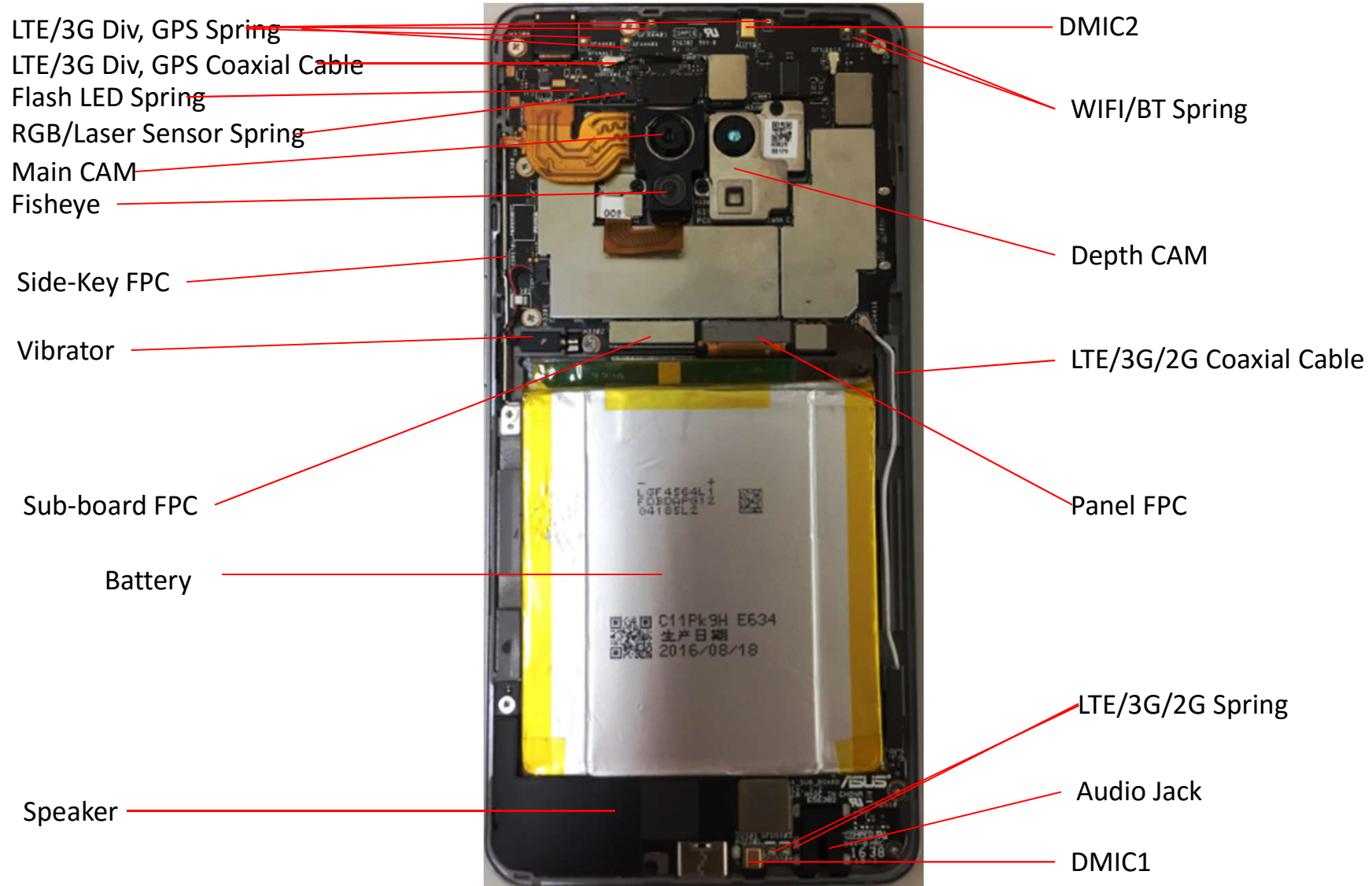
2017/03/30
Winston Lu

System Overview

Device Overview



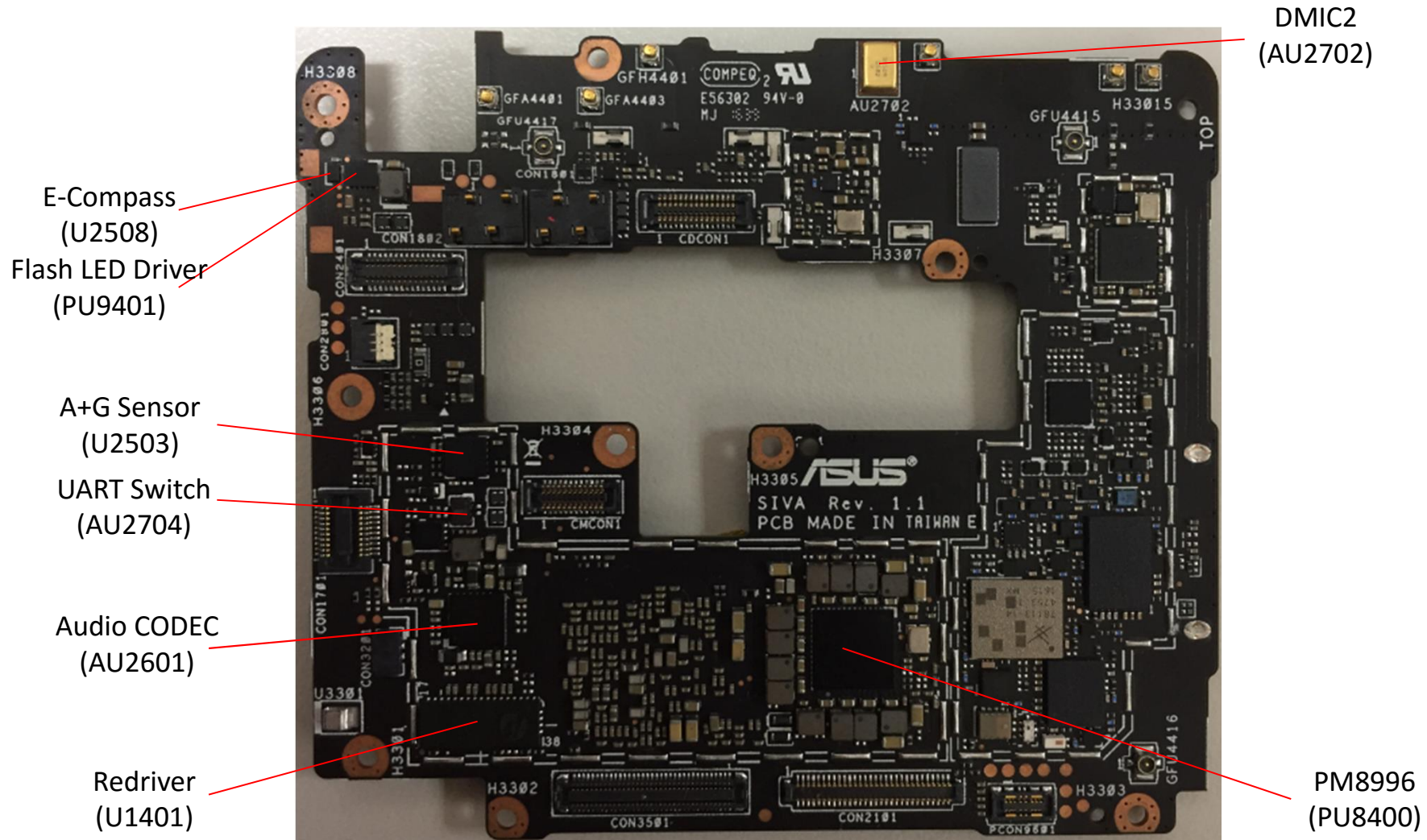
Device Overview



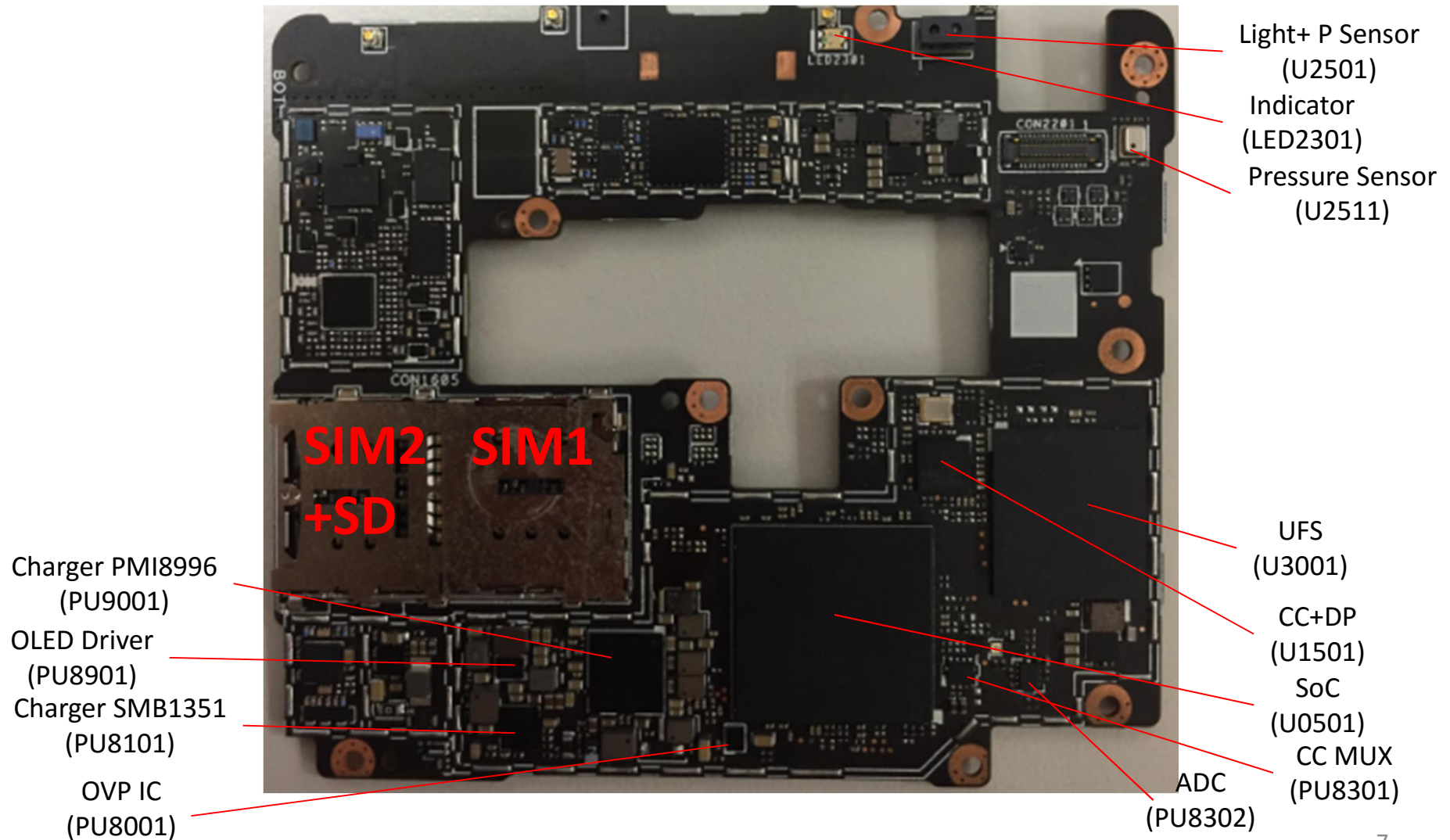
Back Cover



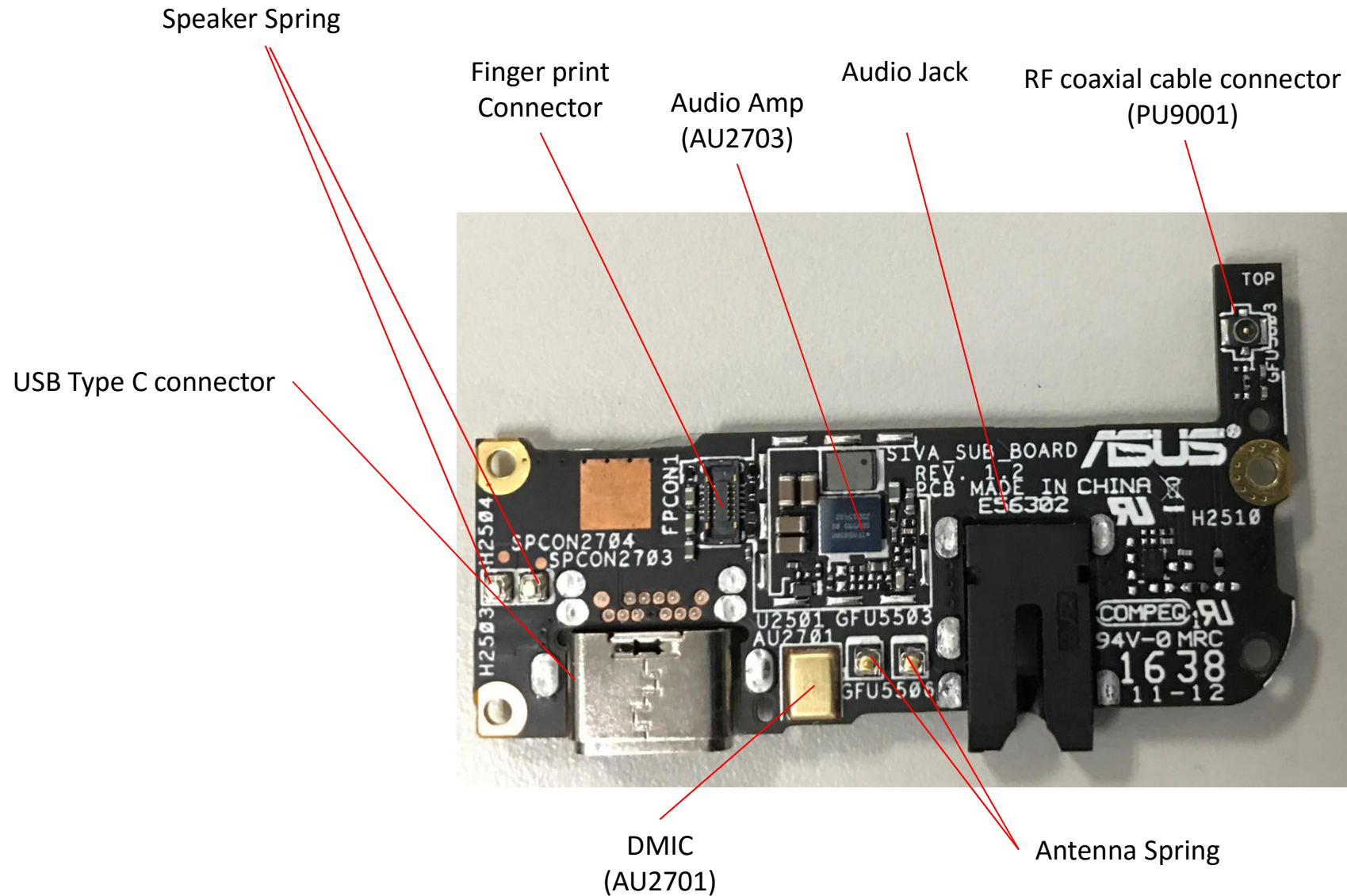
Main Board Overview (Top)



Main Board Overview (Bottom)



Sub Board Overview (Top)

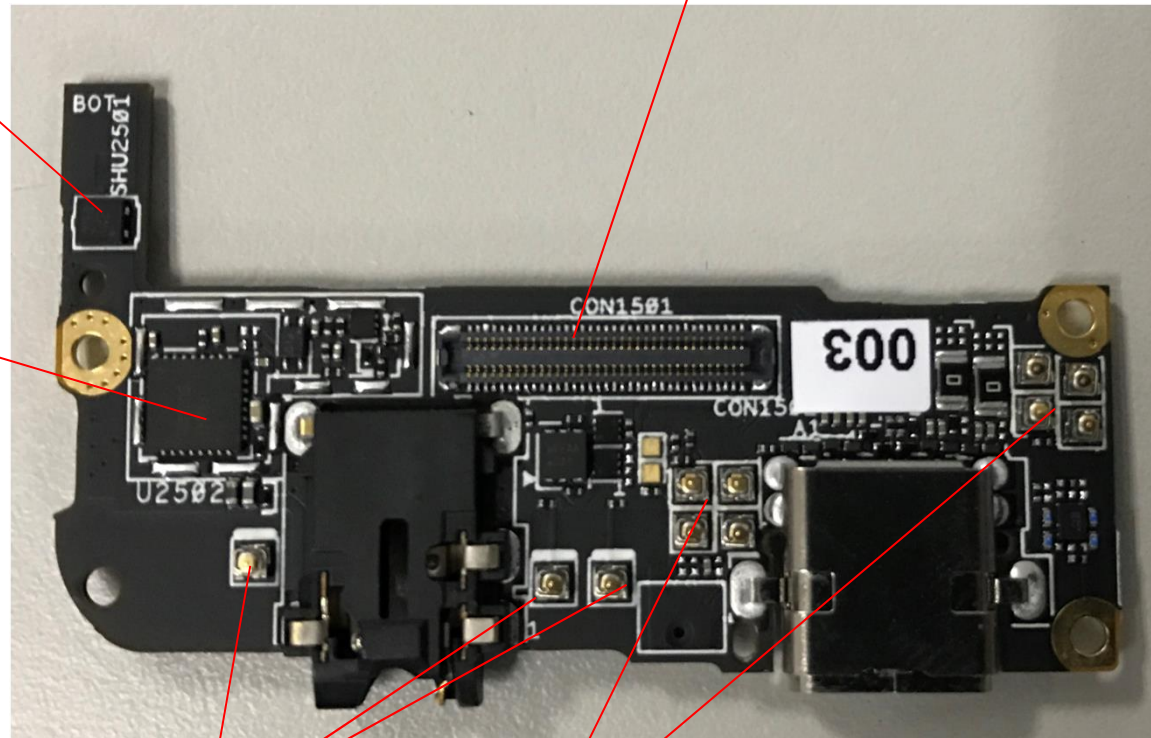


Sub Board Overview (Bottom)

Hall Sensor
(SHU2501)

Sub to MB connector

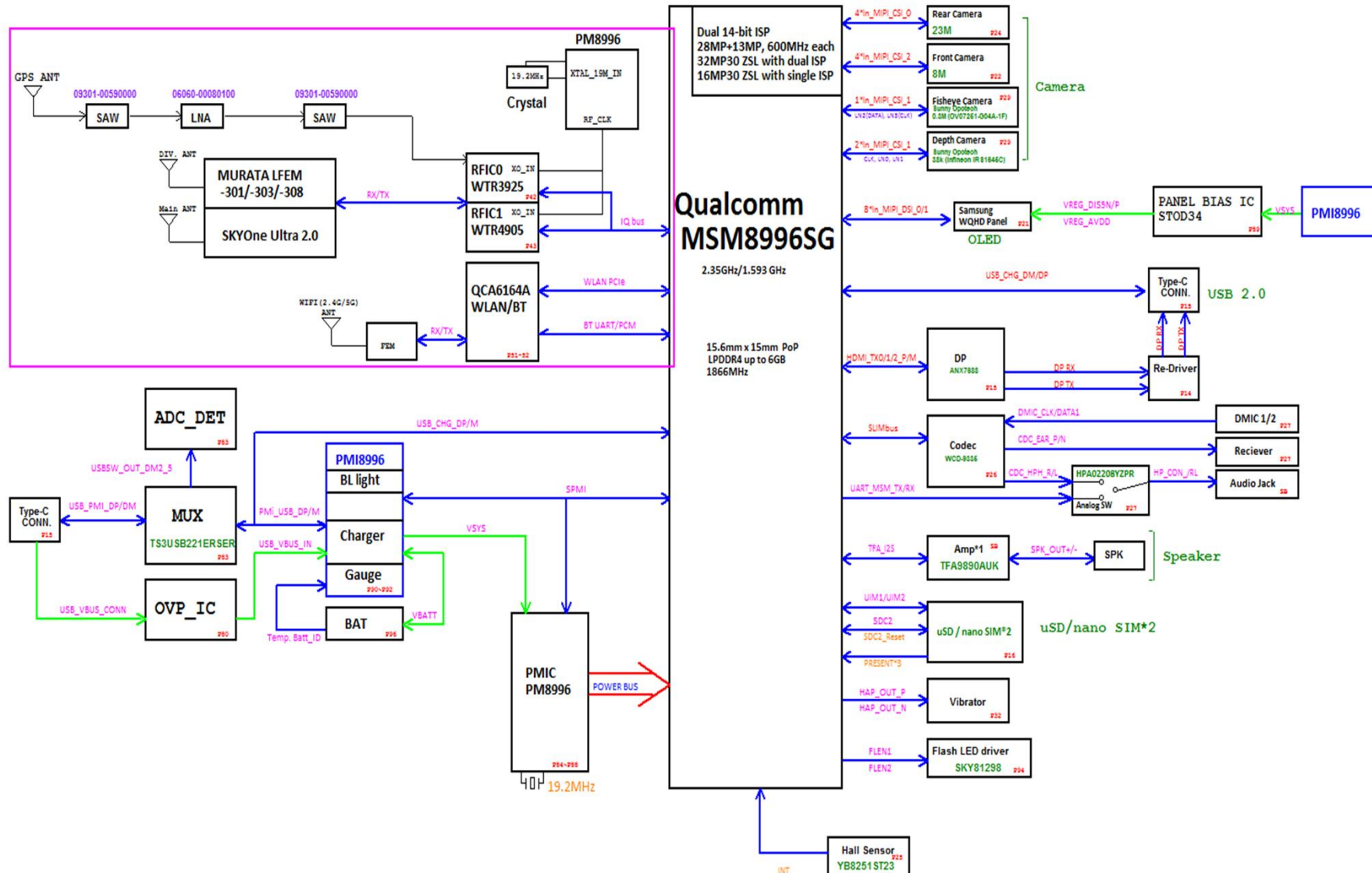
Finger print IC
(U1601)



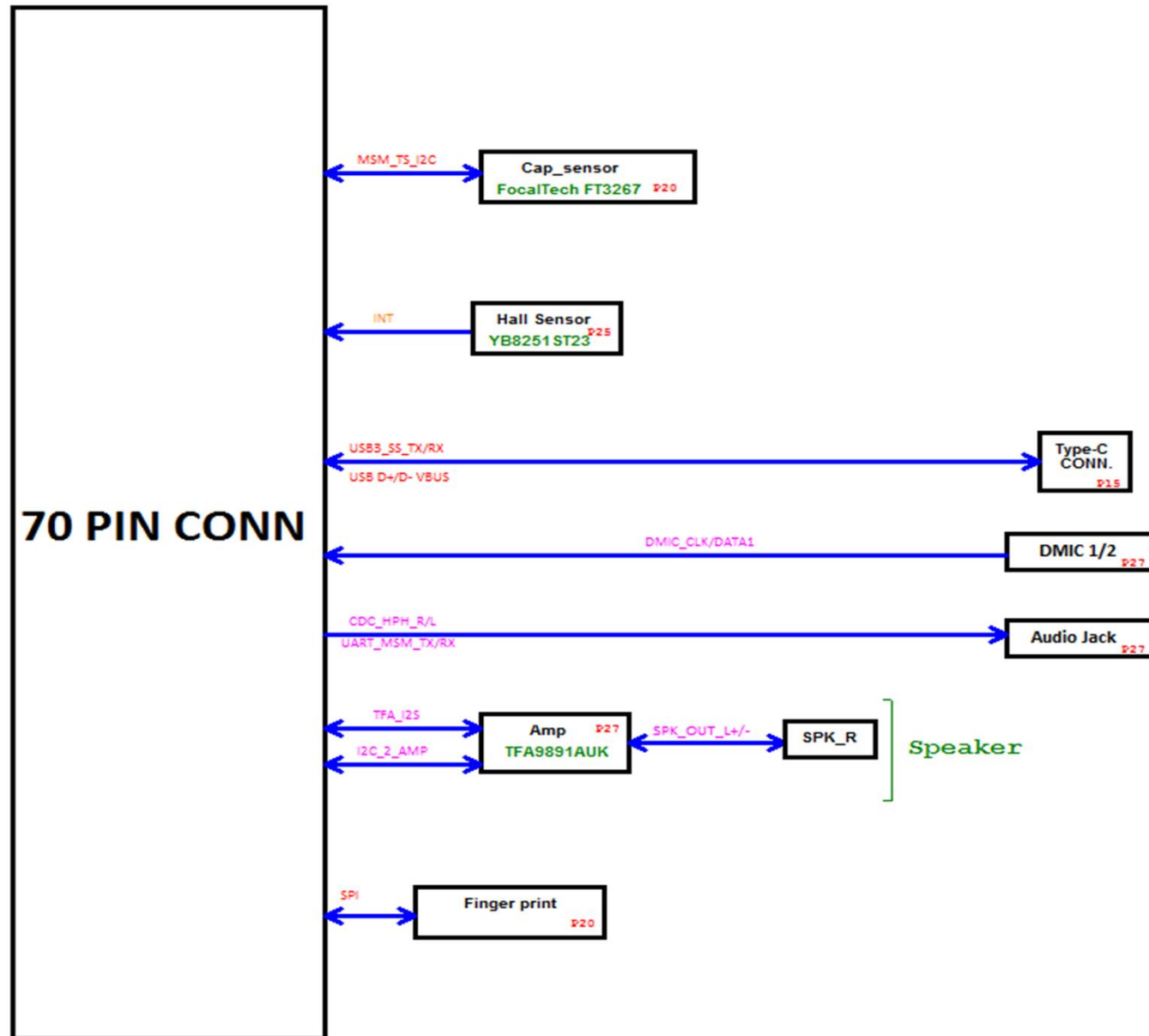
Antenna Spring

Virtual Key Spring

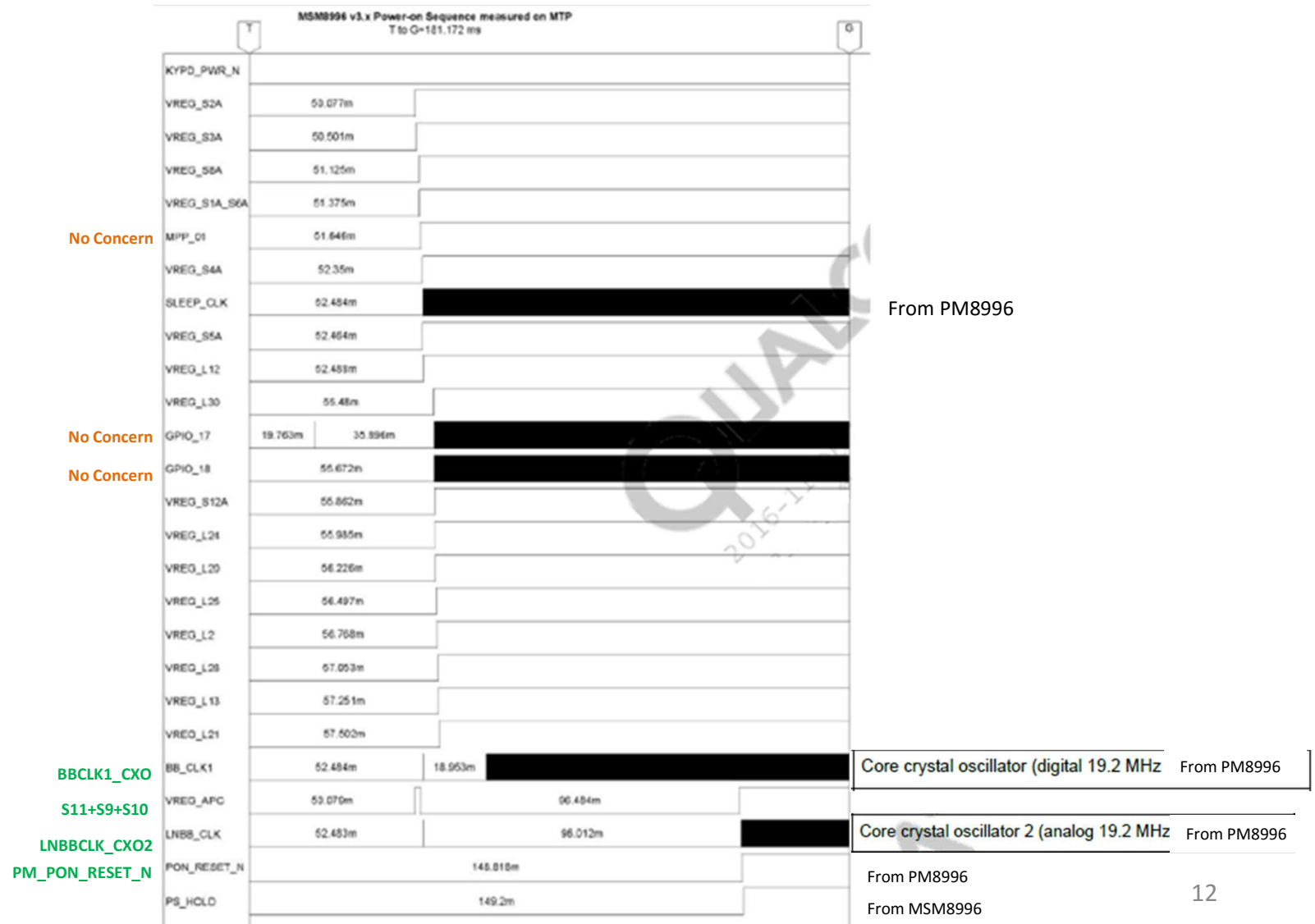
System Function Block Diagram(MB)



System Function Block Diagram(SB)



Power-On Sequence



PMIC Power Rail Voltage

Power Rail	Default (V)	Usage	Operate Voltage(V)
VREG_S2A_0P85	0.85	MSM memory	0.5875-1.1125
VREG_S3A_1P3	1.3	LDOs 2,3,4,11,26,27,28,31	0.375-1.400
VDD_EBI_MSM(S8A)	0.8	MSM APC0/1	0.375-1.1125
VREG_S1A_S6A_0P8	0.8	MSM digital Core	0.375-1.1125
VREG_S4A_1P8	1.8	MSM always ON	1.700-1.950
VREG_S5A_2P15	2.15	Audio buck; LDOs	1.700-2.300
VREG_L12A_1P8	1.8	MSM	1.620-3.600
VREG_L30A_1P8	1.8	Wireless connectivity	1.700-3.600
VREG_S12A_1P15	1.125	LPDDR4	0.5875-1.300
VREG_L24A_3P075	3.075	MSM USB	1.620-3.600
VREG_L20A_3P3	2.95	UFS	1.620-3.600
VREG_L25A_1P2	1.2	UFS	0.950-1.300
VREG_L2A_1P25	1.225	MSM CSI, DSI	0.900-1.300
VREG_L28A_0P925	1.0	MSM PCIe, UFS, PLLs	0.950-1.300
VREG_L13A_2P95	2.95	MSM SDC	1.620-3.600
VREG_L21A_2P95	2.95	SD	1.620-3.600
VREG_APC_0P8	0.8	MSM APC0/1	0.375-1.1125

System Power Tree



Power Tree

Trouble Shooting – MB

- Power Up
- Storage
- Sensors
- Key parts

Power Up

No Boot
USB PHY.
Charging
Gauge
MODEM

Trouble Shooting – No Boot (1/2)

- Check if **battery voltage** (PCON9601) is higher than **3.5V**
- Check if **VSYS** has voltage output
- Check if **PM_PON_RESET_N**, **PS_HOLD** are driven to 1.8V
 - If not, please check each power domain of page 12
- Take X-ray photo with
 - PM8996(PU8400)
 - SoC (U0501)
 - Other devices
- Check if each power domain of PMIC output normally
- Mechanically turn off the device, and check if any power domain of PMIC is short to GND

Trouble Shooting – No Boot (2/2)

- Connect device to PC through USB, check if “QDLoader9008”, present in Device Manager, which means device has entered **Recovery Mode**



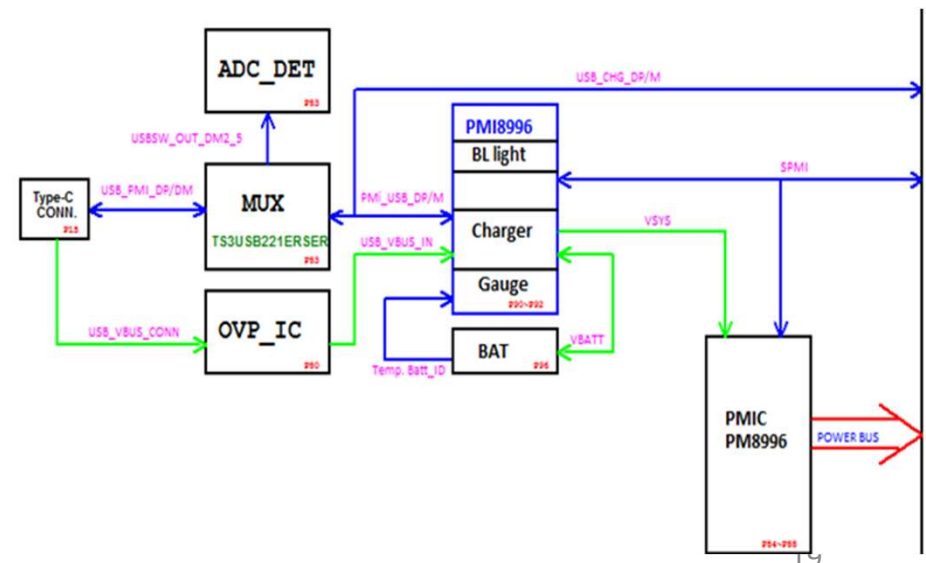
- Try to re-download image

How to enter Recovery Mode:

- Press Volume up + Volume down + Power Key
- Reheat or replace
 - PM8996(PU8400)
 - SoC(U0501)
- Treat it as UFS fail issue

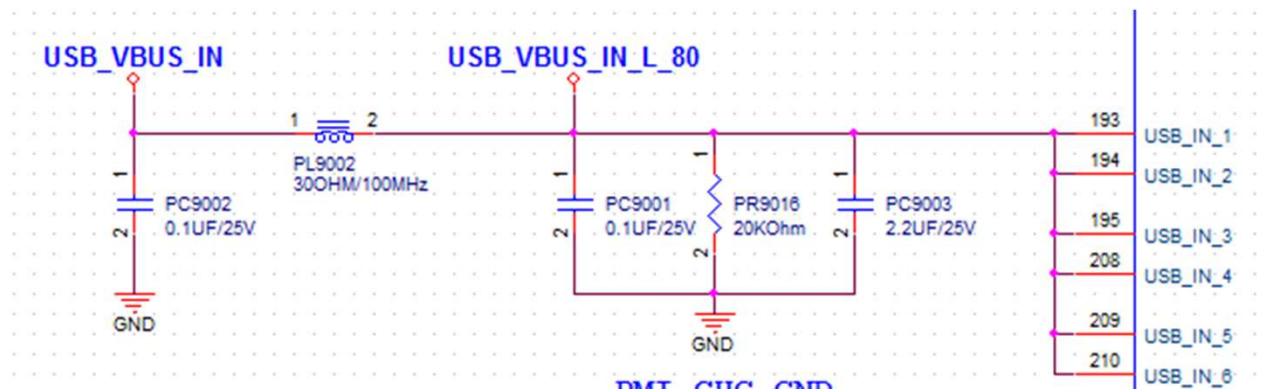
Trouble Shooting – USB Fail : Int. PHY

- Check if **USB_VBUS_CONN** with 5V (PC8001)
- Check if **OVP_IC**(PU8001), **bead**(PL9002) connected well
- Check if **PMi_USB_DP**(PR8314)/ **PMi_USB_DM**(PR8315) have signal on each of them
- Check if **VREG_L24A_3P075**, **VREG_L12A_1P8** with normal on them
- Reheat or replace
 - MUX(PU8301)
 - Charger PMI8996(PU9001)
 - SoC(U0501)



Trouble Shooting – Charging Fail

- Plug USB adapter in, then check if USB_VBUS_IN_L_80 has 5V supplied(**PL9002**)
- Check if battery voltage gradually increases (PCON9601)
- Take X-ray photo with
 - PMI8996 (PU9101)
 - SMB1351 (PU8101)
- Reheat or replace charger PMI8996(PU9001) and SMB1351 (PU8101)



PMI8996

Trouble Shooting – Battery Gauge Fail

- Check if PMIC_SPMI_CLK(C0405), PMIC_SPMI_DATA(C0406) work normal(internal pull-high 1.8V)
- Reheat or replace charger(PU9001, charger with build-in battery gauge)

Trouble Shooting – MODEM Fail

- Check if **VREG_S7A_1P0** ON(1.0V)

Parameter ¹		Min	Max	Unit
VDD_MODEM	Power for modem circuits			
	Turbo	0.795	1.07	V
	Nominal	0.705	0.976	V
	SVS	0.59	0.8	V
	Low SVS	0.515	0.717	V

- Reheat or replace SoC(U0501)

Storage

UFS

SD Card

SIM Card

Trouble Shooting – UFS Fail

- Power :
 - Check **VREG_L20A_3P3** is **3.3V** (C3002)
 - If VREG_L20A_3P3 is 2.95V, check system jump to 9008 mode or not
 - If VREG_L20A_3P3 is 0V, refer to “No Boot” issue
 - Check **WCD_UFS_1V8** is **1.8V** (C3006)
 - Check **VREG_L25A_1P2** is **1.2V** (C0954)
- Interface :
 - Check UFS_RESET is 1.2V (T0411)
 - Check UFS interface works normally(UFS_REF_CLK:C0402; UFS_RX_M:T0410; UFS_RX_P:T0409; UFS_TX_M:T0408; UFS_TX_P:T0407)
 - Check R0407 connected to GND
- Take X-ray photo with UFS and SoC
- Reheat or replace UFS(U3001)
- Reheat or replace SoC(U0501)

Trouble Shooting – SD Card Fail

- Check if SMT issue
- Insert SD card
 - Detect : check if SD_CARD_DET pulled high(R1616)
 - Power :
 - Check VREG_L13A_2P95 is 1.8V(UHS card) or 2.95V (HS, NS card)(C0953)
 - Check VREG_L21A_2P95 is 2.95V (C2301) – CON1605
 - Clock : Check if SDC2_CLK with signal (R1608)- SoC
 - Interface : Check SDC2 interface works normally
- Reheat or replace CON1605
- Take X-ray photo with SoC
- Reheat or replace SoC(U0501)

Trouble Shooting – SIM Card Fail

- Check if SMT issue
- Check if SIM card abnormal itself
- Insert SIM Card
 - Detect : Check UIM1_PRESENT/UIM2_PRESENT pulled high (R1609)
 - SIM1 Power : Check if **VREG_L9A_UIM1** is 3.3V or 1.8V for **SIM1**
 - SIM2 Power : Check if **VREG_L10A_1P8** is 3.3V or 1.8V for **SIM2**
 - Interface : Check if UIM1/ UIM2 interface works normally
- Reheat or replace CON1605
- Take X-ray photo with SoC
- Reheat or replace SoC(U0501)

Sensors

Laser & RGB

Light & Proximity

Pressure

Acceleration & Gyro

e-Compass

Trouble Shooting – Laser or RGB Fail

- Check the module FPC **connected** to MB well
- Power :
 - Check if **PW_ADC_2V8** is **2.8V** (C1801), which power is always on
- Reheat or replace PU9503
- Check I2C Bus (MSM_I2C_SDA_LSR_RGB) :
 - Command query each other
 - Check if the I2C interface MSM_I2C_SDA_LSR_RGB(D1802), MSM_I2C_SCL_LSR_RGB (D1801) work normally
- Replace the module FPC
- Take X-ray photo with SoC
- Reheat or replace SoC(U0501) if interface is abnormal

Trouble Shooting – Light or Proximity Fail

- Check Light Sensor itself :
 - Check Rubber and U2501 polarization
 - Wipe and Re-assemble
- Power :
 - Check if **VREG_L19A_3P3** is **3.3V** (C2514)
 - Check if **VREG_LVS2A_1P8** is **1.8V** (R2501)
- Check I2C Bus (SSC_I2C_3_SCL) :
 - Command query other sensor
 - Check if the I2C interface SSC_I2C_3_SDA (R3121), SSC_I2C_3_SDA (R3122) work normally
- Take X-ray photo with Sensor, PM8996, and SoC
- Reheat or replace U2501
- Reheat or replace PM8996 (PU8400) if power is abnormal
- Reheat or replace SoC (U0501) if interface is abnormal

Trouble Shooting – Pressure Fail

- Power :
 - Check if **VREG_LVS2A_1P8** is **1.8V** (R2501)
- Check I2C Bus (SSC_I2C_3_SCL) :
 - Command query other sensor
 - Check if the I2C interface SSC_I2C_3_SDA (R3121), SSC_I2C_3_SDA (R3122) work normally
- Take X-ray photo with Sensor, PM8996, and SoC
- Sensor :
 - Check polarization
 - Reheat or replace U2511
- Reheat or replace PM8996(PU8400) if power is abnormal
- Reheat or replace SoC(U0501) if interface is abnormal

Trouble Shooting – Acceleration & Gyro Fail

- Power :
 - Check if **VREG_LVS2A_1P8** is **1.8V** (R2501)
- Check I2C Bus (SSC_I2C_3_SCL) :
 - Command query other sensor
 - Check if the I2C interface SSC_I2C_3_SDA (R3121), SSC_I2C_3_SDA(R3122) work normally
- Take X-ray photo with Sensor, PM8996, and SoC
- Sensor :
 - Check polarization
 - Reheat or replace U2503
- Reheat or replace PM8996(PU8400) if power is abnormal
- Reheat or replace SoC(U0501) if interface is abnormal

Trouble Shooting – e-Compass Fail

- Power :
 - Check if **VREG_LVS2A_1P8** is **1.8V** (R2501)
- Check I2C Bus (SSC_I2C_3_SCL) :
 - Command query other sensor
 - Check if the I2C interface SSC_I2C_3_SDA (R3121), SSC_I2C_3_SDA(R3122) work normally
- Take X-ray photo with Sensor, PM8996, and SoC
- Sensor :
 - Check polarization
 - Reheat or replace U2508
- Reheat or replace PM8996(PU8400) if power is abnormal
- Reheat or replace SoC(U0501) if interface is abnormal

Key part Function

Touch

Display

Main CAM

Front CAM

Depth CAM

Fisheye

Audio- DMIC2, Receiver

Vibrator

Side-key

Trouble Shooting – Touch Fail

- Check Panel FPC 60pin connector(**CON2101**) to TP Sensor is properly **assembled**
- Swap : check sensor is functional
- Power :
 - Check **VREG_L22A_3P3** has 3.3V supplied(PC8815)
 - Check **VREG_L14A_1P8** has 1.8V supplied(PC8813)
- I2C :
 - Check MSM_TS_I2C_SCL/SDA works normally (R3129/R3130)
- GPIO :
 - Check INT(R2101) & works normally
- Take X-ray photo with PM8996, and SoC
- Reheat or replace PM8996(PU8400) if power is abnormal
- Reheat or replace SoC(U0501)

Trouble Shooting – No Display

- Check Panel FPC 60pin connector (**CON2101**) to panel is properly assembled
- Swap : Replace Panel & MB cross certification
- Power :
 - Check **VREG_L22A_3P3** has **3.3V** supplied(PC8815) from PM8996
 - Check **VREG_L14A_1P8** has **1.8V** supplied(PC8813) from PM8996
 - Check **VREG_DIS5P** is **4.6V** (C2102) from STOD34 (PU8901)
 - Check **VREG_DIS5N** is **-4V** (C2103) from STOD34 (PU8901)
 - Check **VREG_AVDD** is **7.6V** (C2101) from STOD34 (PU8901)
- Signal : Check MIPI interface has driven to HS voltage (150~250mV).
 - If not check SOC / connector
- Take X-ray photos
 - driver STOD34 (PU8901) or PM8996 if power is abnormal
 - SoC(U0501) /
- Reheat or replace
 - connector(con2101) /
 - driver STOD34 (PU8901) or PM8996 if power is abnormal
 - SoC(U0501) /

Trouble Shooting – Main CAM Fail

- Check Main Camera Module is properly **assembled**
- Swap : Check Main Camera & MB is functional by cross certification
- Check I2C Bus :
 - Command query **Front CAM**
 - Check CCI_I2C_SDA0/SCL0 works normally (C2412 & C2411)
- Power :
 - Check **V_1P80_DOVDD** has 1.8V supply (C2404) from PMIC VREG_LVS1A_1P8 (**supply for all camera**)
 - Check **V_2V8_CAM_CB** has 2.8V supply (C2402) from PU9501
 - Check Enable (CB_CAM_2V8_EN @ R9501) is normal
 - Check **V_1P0_DVDD** has 1.0V supply (C2407) from PM899 VREG_L3A_1P0
 - Check **V_2P80_VCM** has 2.8V supply (C2408) from PU9510
 - Check **V_2P80_OIS** has 2.8V supply (C2410) from PM8996 VREG_L23A_2P8
- GPIO :
 - Check MAIN_CAM_RST has pull-down to 0V (C2409)
 - Check CB_CAM_2V8_EN has driven to 1.8V (R9501)
- Main Clock :
 - Check MAIN_CAM_MCLK 19.2MHz clock (C0505)
- MIPI :
 - Check MIPI interface has driven to HS voltage (330mV)
 - Check MIPI CLK has driven to 1.8V
- X-ray, Reheat or replace
 - connector(CON2401)
 - PU9501 if power is abnormal
 - PU9510 if power is abnormal
 - PMIC(PU8400) if power is abnormal
 - SoC(PU0501) if power is abnormal

Trouble Shooting – Front CAM Fail

- Check Front Camera Module is properly assembled
- Swap : Check Front Camera & MB is functional by cross certification
- Check I2C Bus
 - Command query main CAM
 - Check CCI_I2C_SDA0/SCL0 works normally (C2209 & C2210)
- Power :
 - Check SUB_CAM_1V8 has 1.8V supply (C2203) from PM8996 VREG_LVS1A_1P8 (supply for all camera)
 - Check SUB_CAM_2V8_SEN has 2.8V supply (C2204) from PU9504
 - Check CF_CAM_2V8_EN has driven to 1.8V (R9502)
 - Check SUB_CAM_1V2 has 1.2V supply (C2205) PM8996 VREG_L27A_1P0 (dedicate for front cam)
- GPIO :
 - Check SUB_CAM_PWDN# has pull-up to 1.8V (C2207)
- Main Clock :
 - Check SUB_CAM_MCLK1_R 19.2MHz clock (C0506)
- MIPI :
 - Check MIPI interface has driven to HS voltage (330mV)
 - IF not check SoC / connector
- X-ray, Reheat or replace
 - connector(CON2201)
 - PU9504/
 - PMIC(PU8400)/
 - SoC(U0501)/

Trouble Shooting – Depth CAM Fail

- Check Depth Camera Module is properly **assembled**
- Swap : Check Depth Camera & MB is functional by cross certification
- **Power** :
 - Check **CAM_1V5** has **1.5V** supply (CDC6) from PU8207
 - Check **VREG_LVS1A_1P8** has **1.8V** supply (CDC4) from PMIC (**supply for all camera**)
 - Check **+VCSEL_3V3** has **3.3V** supply (CDC8) from PU8203
 - Check **+CD_3V3** has **3.3V** supply (CDC15) from PU9508 and PU8204
 - Enable by DEPTH_LDO_EN (PR9501)
- Check I2C Bus
 - Command query **fish eye**
 - Check CCI_I2C_SDA1/SCL1 works normally (CDC11 & CDC12)
- GPIO :
 - Check **DEPTH_LDO_EN** has pull-up to **1.8V** (PR9501)
 - Check **CD_CAM_RESET_N** has pull-up to **1.8V** (CDC10)
 - Check **CD_CAM_IMG_START_CON** has driven to **1.8V** (CDC14)
- Main Clock
 - Check CD_CAM_MCLK_R 24MHz clock (CDC9)
- MIPI
 - Check MIPI interface has driven to HS voltage (330mV)
 - IF not check SoC / connector
- X-ray, Reheat or replace
 - Connector(CMCON1)
 - PU8207/
 - PU8203/
 - PU8204
 - PU9508/
 - PMIC(PU8400)/
 - SoC(U0501)/

Trouble Shooting – Fisheye Fail

- Check Fisheye Module is properly assembled
- Swap : Check Fisheye & MB is functional by cross certification
- Power :
 - AVDD : Check VREG_L29A_2P8 has 2.8V supply (CMC2) from PM8996
 - VDDIO : Check VREG_LVS1A_1P8 has 1.8V supply (CMC4) (supply for all camera)
- Check I2C Bus :
 - Command query Depth camera
 - Check CCI_I2C_SDA1/SCL1 works normally (CDC11 & CDC12)
- GPIO :
 - Check CM_CAM_XSHUTDOWN_CON has pull-up to 1.8V (CMC6)
- Main Clock
 - Check CFE_CAM_MCLK_R 24MHz clock (CMC5)
- MIPI :
 - Check MIPI interface has driven to HS voltage (330mV),
 - if not check SoC / connector
- X-ray, Reheat, or replace
 - Connector(CMCON1)
 - PMIC(PU8400)/
 - SoC(U0501)/

Trouble Shooting – DMIC2 Fail

- Check rubber
- Re-assemble
- Check Power :
 - Check if MIC_BIAS3 with 1.8V(AC2752)
- Check Signal :
 - Check if DMIC2_DATA, DMIC2_CLK have signal(AC2753, AC2750)
- X-ray, Reheat ,or replace
 - AU2702
 - CODEC(AU2601)
 - Reheat or replace SoC(U0501)

Trouble Shooting – Receiver Fail

- **Swap** :
 - Check Receiver & MB is functional by cross certification
- **Spring** :
 - Check Receiver Spring is not damaged
- Check if REC_P, REC_M have signal(AH2701, AH2702)
- Reheat or replace
 - CODEC(AU2601)
 - Reheat or replace SoC(U0501)

Trouble Shooting –Vibrator Fail

- Check HAP_OUT_P_R/ HAP_OUT_N_R with signal (T3201,T3202)
- Reheat or Replace CON3201 & PMI8996(PU9001)

Trouble Shooting – Side Key Fail

- Check Side-Key FPC is properly **assembled**
- **Replace** Side-Key FPC and check if problem fixed
- **Press** key pad and check :
 - VOL_UP_N(T2802)/ PM_RESIN_N(T2803)/ PHONE_ON_N(T2801)
pulled **low**
- **Release** Key pad and check :
 - VOL_UP_N(T2802)/ PM_RESIN_N(T2803)/ PHONE_ON_N(T2801)
drives to **1.8V**
- Reheat PM8996(PM8401) and PMI8996(PU9001)

Trouble Shooting – SB

- Sensors
- Key parts

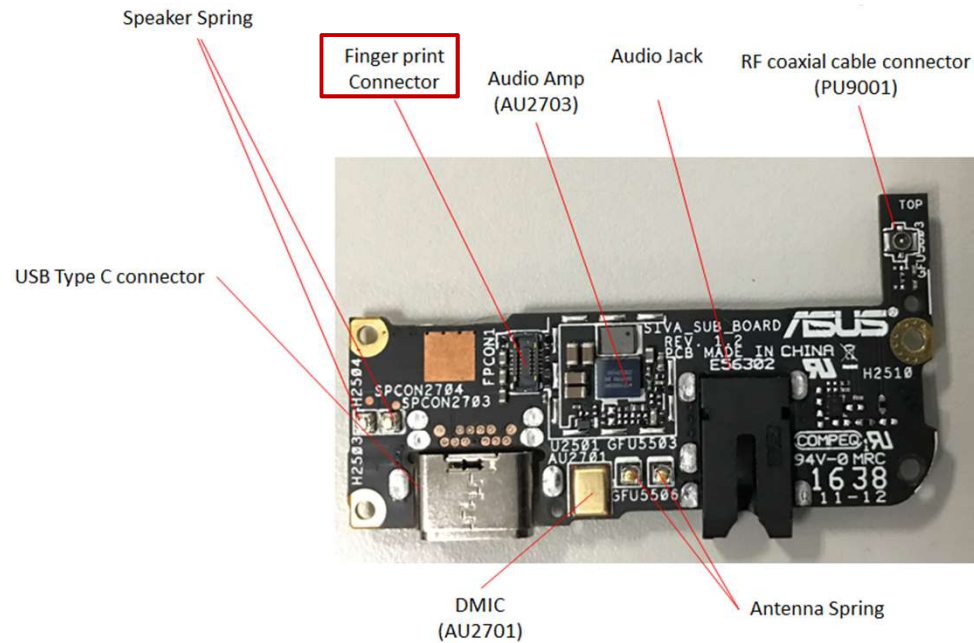
Trouble Shooting – SB

- Sensors
- Key parts

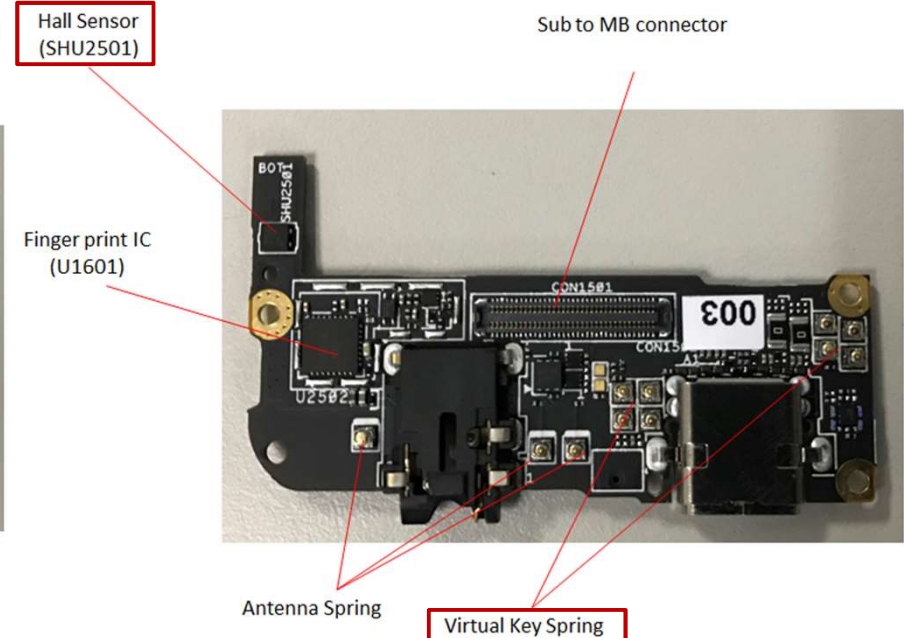
Sensors

- Hall Sensor
- Cap sensor(Virtual Key)
- Finger printer

Top

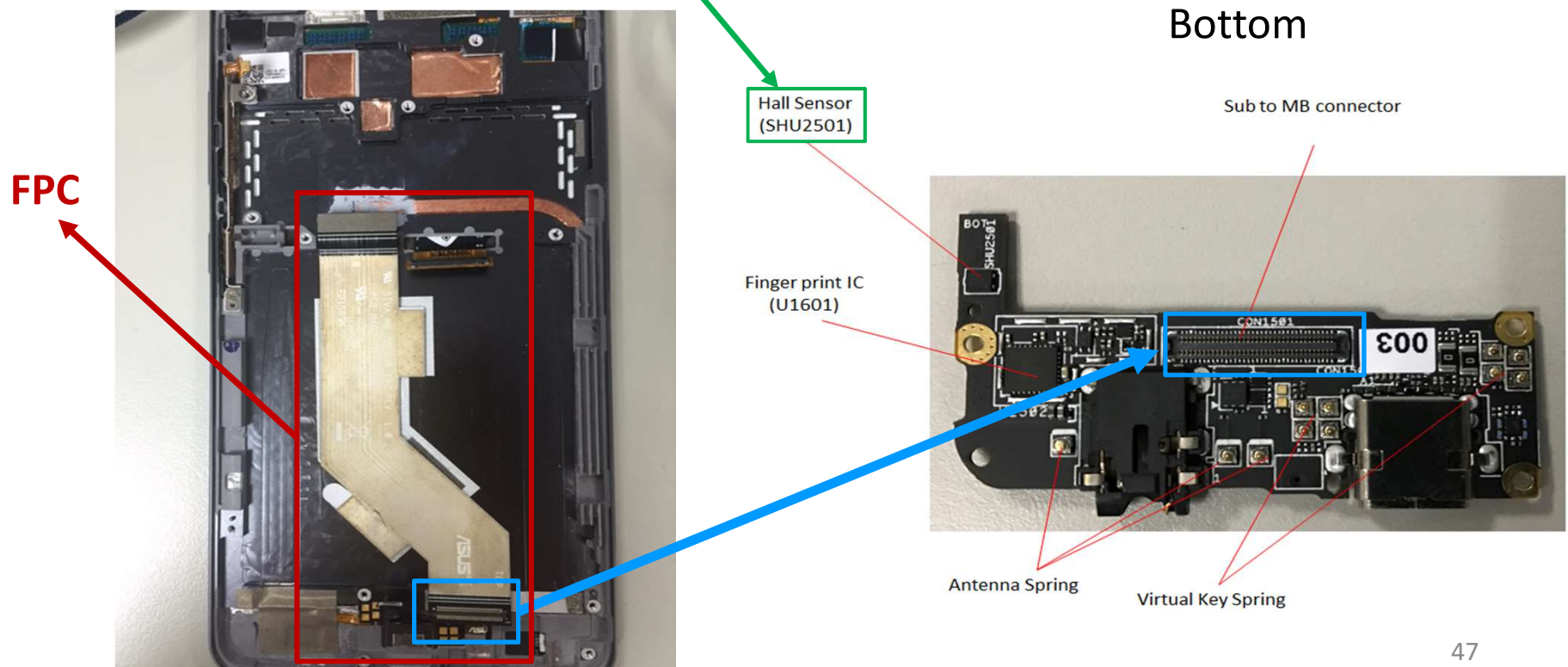


Bottom



Trouble Shooting – Hall sensor Fail

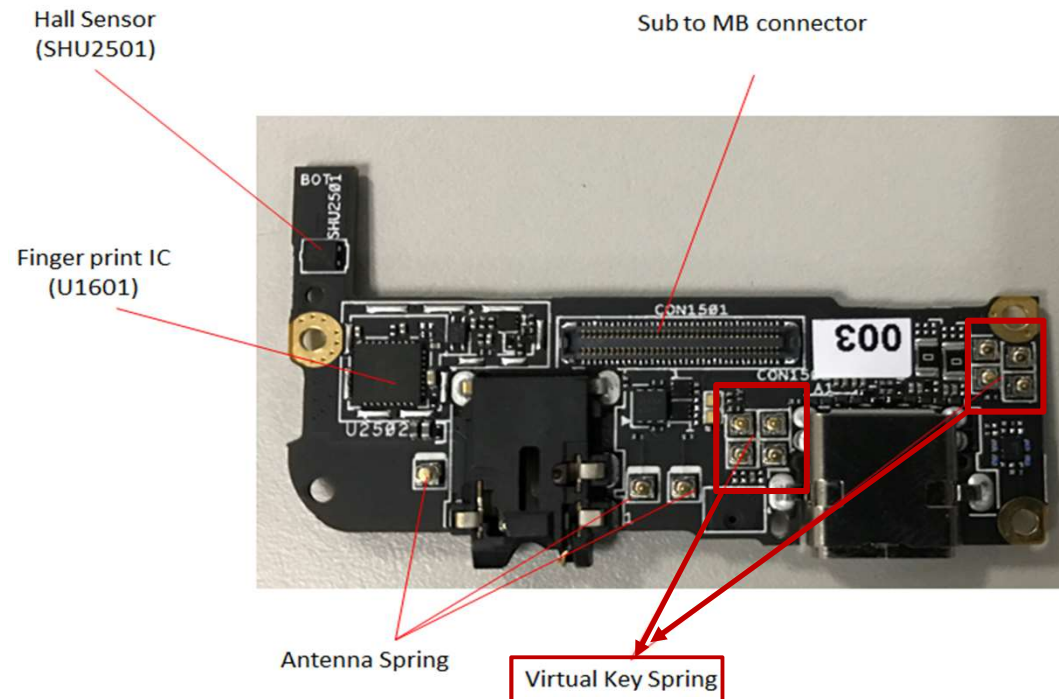
- Check if **PW_SB_1V8** is 1.8V
- Check **SB** to **Main FPC** is properly assembled
- Reheat or replace **Hall sensor (SHU2501)**



Trouble Shooting – Cap sensor Fail

- Check if **virtual key spring** is not damaged
- Check if FP_VCC_3V3 is **3.3V**
- Check if VK_IOVCC is **1.8V**
- **Check SB to Main FPC is properly assembled**
- Reheat or replace **Cap sensor control IC (FT3267)**

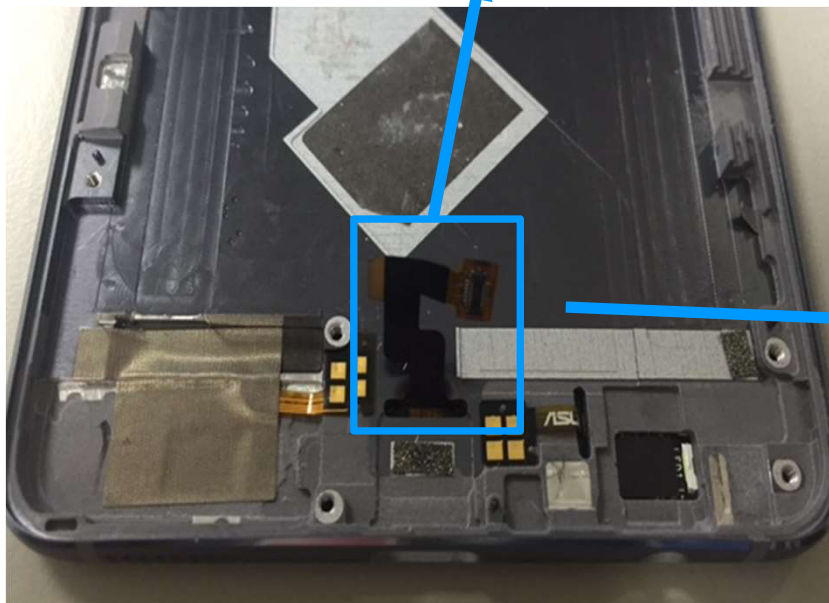
Bottom



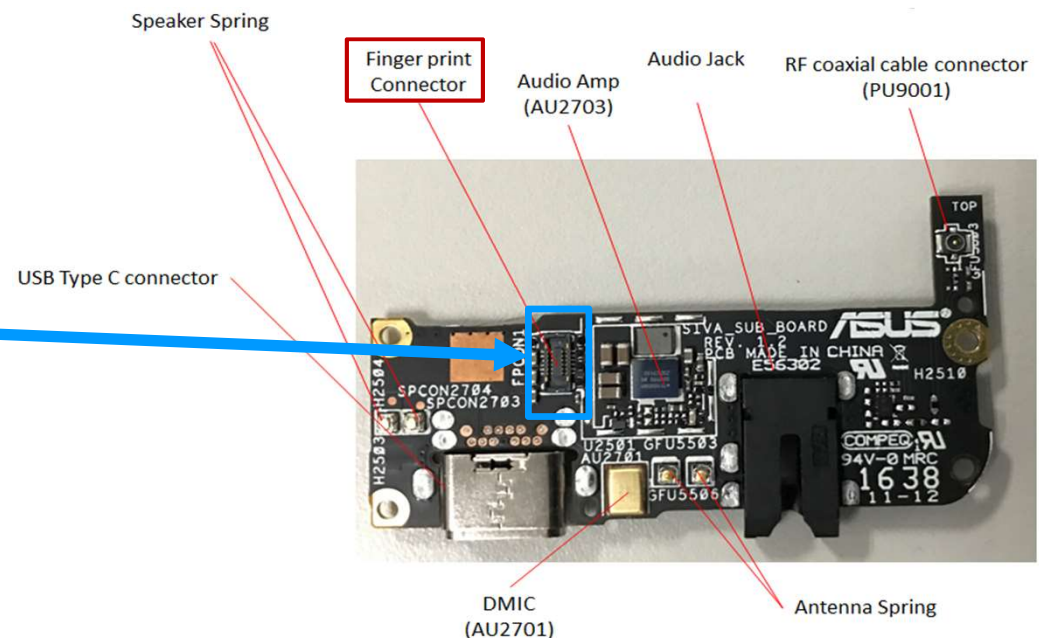
Trouble Shooting – Finger Printer

- Check finger printer FPC is properly assembled
- Check SB to Main FPC is properly assembled
- Check FP_VCC_3V3 is 3.3V and FP_VDDIO_1V8 is 1.8V
- Change Finger printer component

Finger printer



Top

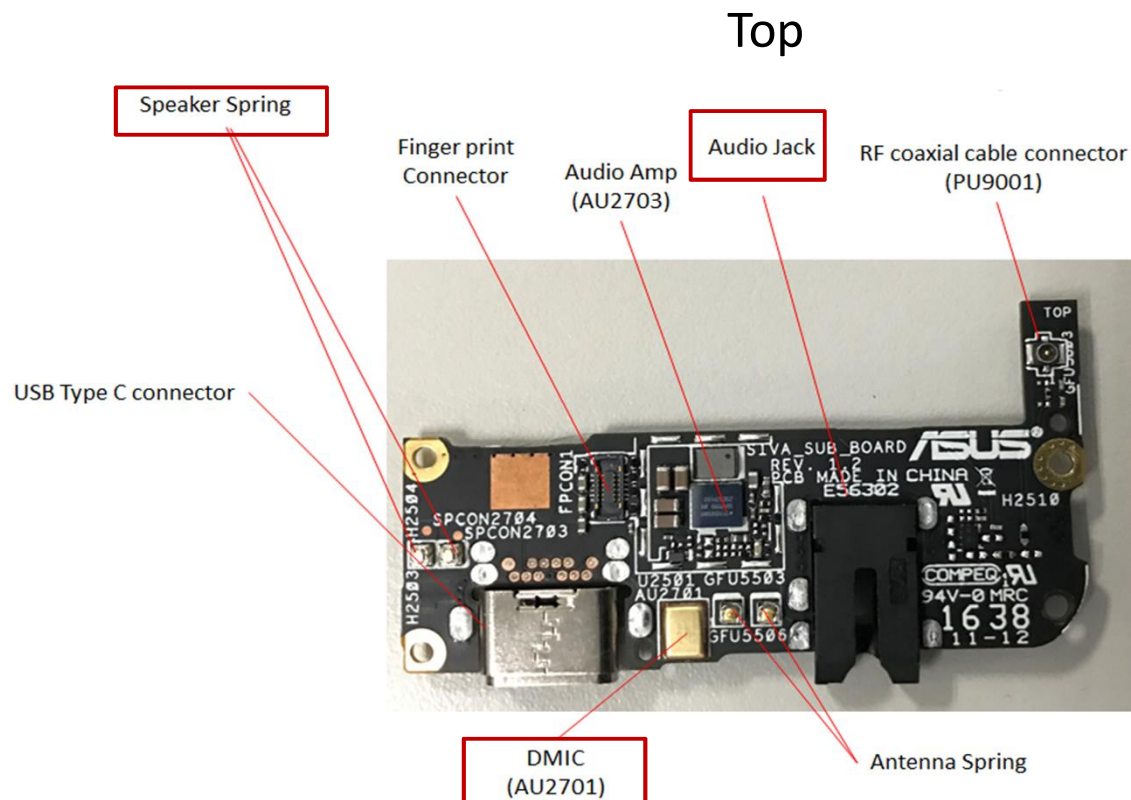


Trouble Shooting – SB

- Sensors
- Key parts

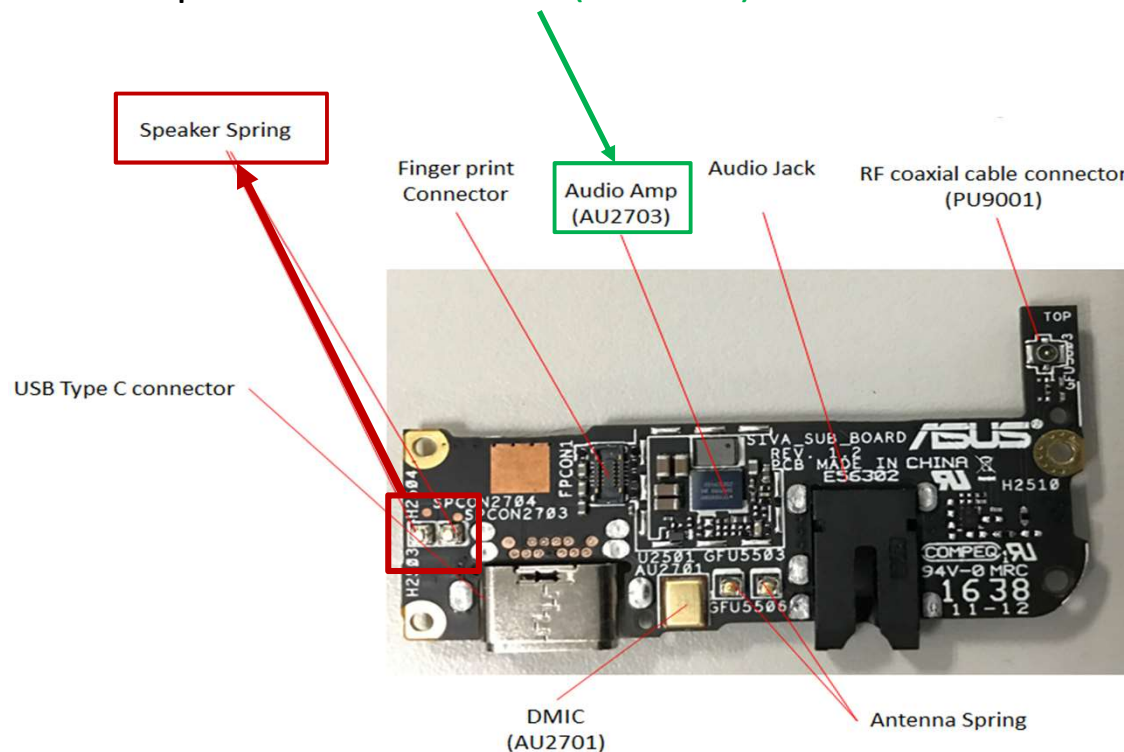
Key parts

- Audio Speaker
- DMIC1
- Headset (Audio Jack)



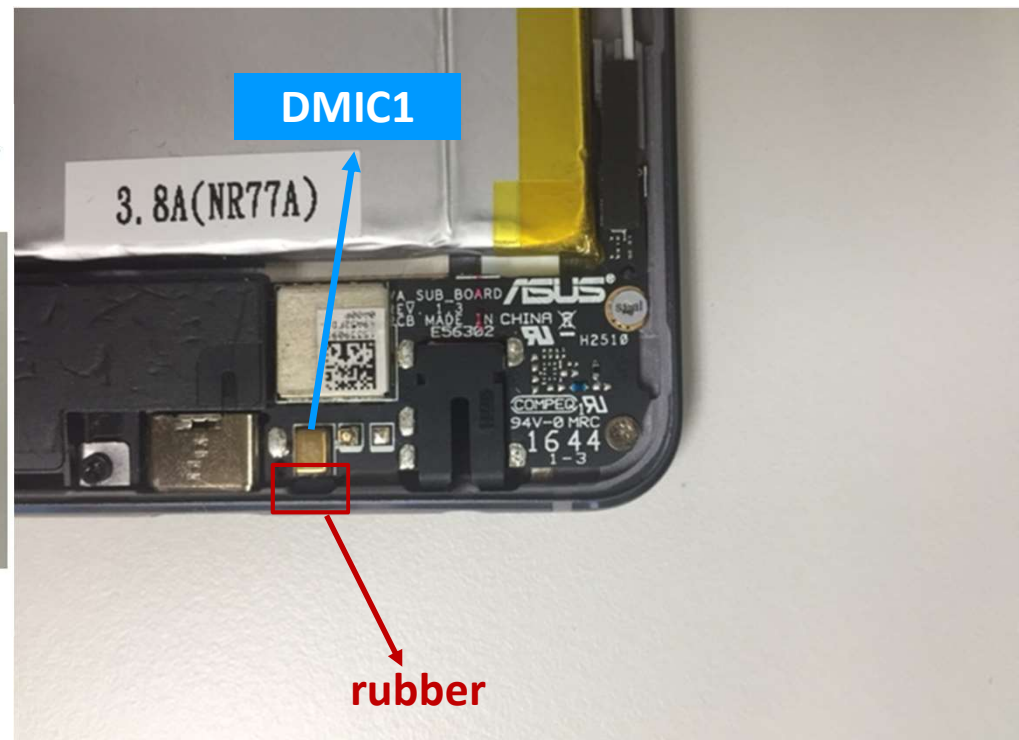
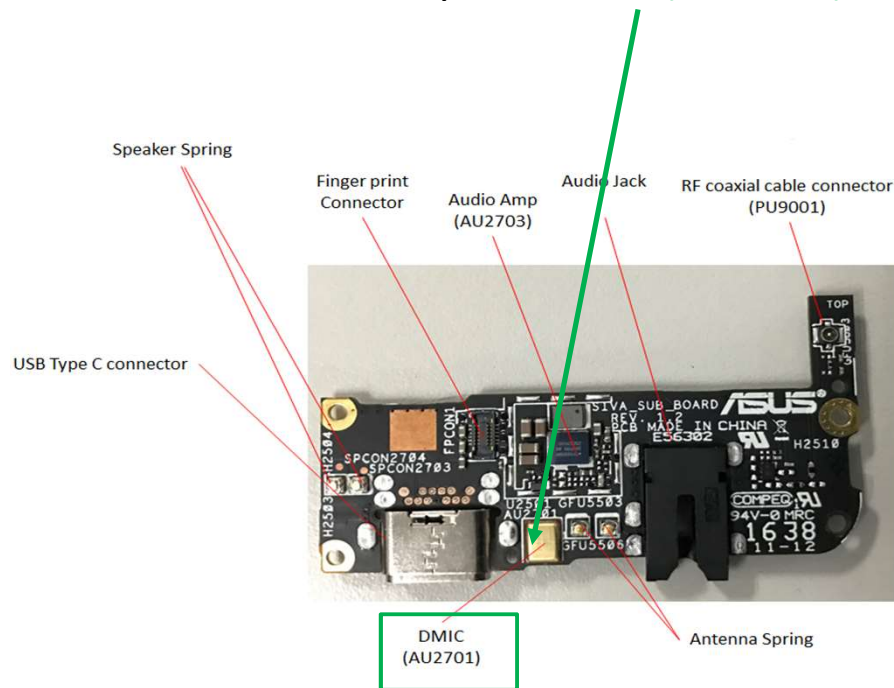
Trouble Shooting – Audio Speaker

- Check speaker is properly assembled
- Check if **Speaker spring is not damaged**
- Check if **PW_SB_1V8** and **VSYS** on SB is OK
- **Check SB to Main FPC is properly assembled**
- Reheat or replace **Audio AMP IC (AU2703)**



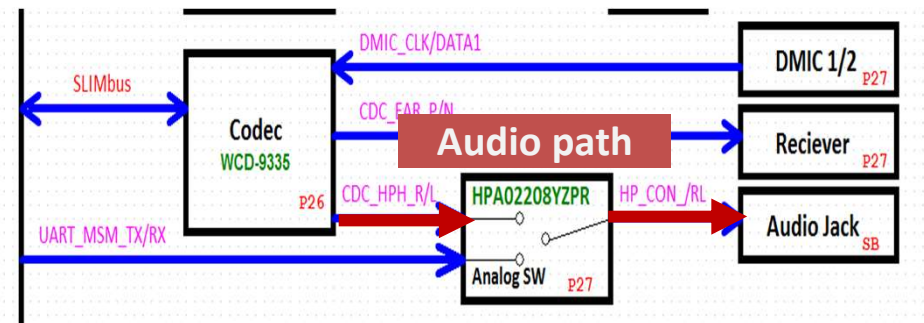
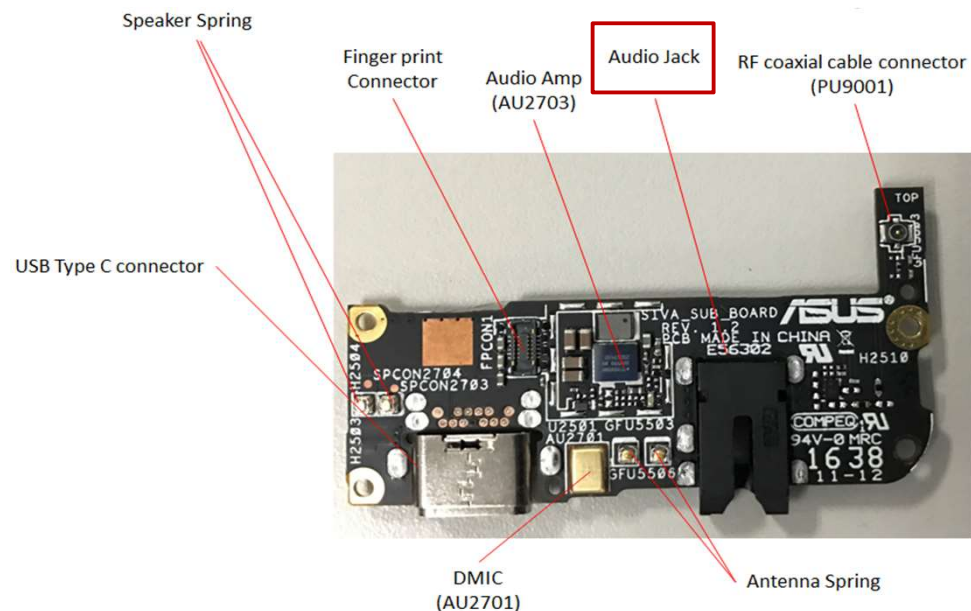
Trouble Shooting – DMIC1

- Check **rubber** is setting OK by **DMIC1**
- Check SB to Main FPC is properly assembled
- Check **MIC_BIAS3** is 1.8V
- Check **DMIC1_CLK** is OK
- Reheat or replace **DMIC(AU2701)**



Trouble Shooting – headset(Audio Jack)

- Check if SW is set to **Audio path** not UART
- **Check SB to Main FPC is properly assembled**
- Check if **MIC_BIAS2** is 1.8V



Thank You